[8] Seat No.: ______

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SARDAR PATEL UNIVERSITY
BCA SEM-I EXAMINATION 2017 (NC)
Digital Computer Electronics (US01EBCA01)

Time: 2:00 p.m. to 4:00 p.m.
Date: 17/03/2017, Friday

Total Marks: 70

Q.1 Multiple Choice Questions **[10]**

1 The _______ gate has two or more input signals. All inputs must be high to get a high output.
[A. AND B. OR C. NAND D. NOR]

2 De Morgan's first theorem says that a NOR gate is equivalent to a _______________.
[A. bubbled OR B. bubbled NOR C. bubbled AND D. AND bubbled]

3 A ______ is a combinational circuit that converts binary information from the 2n coded inputs to an n outputs.
[A. Half Adder B. Decoder C. Encoder D. Comparator]

4 **In k-map, pair eliminates _______ variable.**
[A. one B. two C. three D. four

5 In Comparator, _______ gate is use for comparing bits in word.
[A. AND B. XOR C. NOR D. XNOR]

6 A combinational circuit that performs the arithmetic addition of two bits is called_____.
[A. Full Adder B. Binary Adder C. Half Adder D. Decoder]

7 Half adder consist of________ & _______ Gates.
[A. XOR, AND B. XOR, OR C. XNOR, AND D. XNOR,OR]

8 In D flip-flop, when CLK is low then input is _______.
[A. high B. Don't care C. low D. Not change]

9 A register is a group of ______ that work together as a unit.
[A. multiplexer B. decoder C. flip-flop D. gates]

10 A multiplexer is also called a ________.
[A. data multiplier B. data selector C. data inverter D. data remover]

Q.2 Write Short Questions. [Any 10] **[20]**

1. Prepare truth table for: $\overline{A}B + \overline{B}C$
2. Write **Distributive Law** for addition and multiplication.
3. Prove that **NAND Gate** is equivalent to **Bubbled OR Gate** using Truth table.
4. Draw **K-Map for 3 variables** and randomly show **pair and quade** in same.
5. Draw circuit of **Comparator**.
6. Explain **Sum of Product**.
7. Draw circuit of **Half Adder**.
8. Draw Truth table for 3 input **Full Adder** showing carry and sum.
9. Write a note on **Multiplexer** without circuit.
10. Draw circuit and truth table for **unclocked D Flip-Flop**.
11. Differentiate between **Shift Left** and **Shift Right Register** with proper example.
12. Write a note on **Controlled Buffer Register**.

Q.3(A) Write a detail note on following gates: **AND, NAND, Bubbled AND** **[06]**

(B) Prove that $AB\overline{C}+ABC=AB$ using truth table. **[04]**

OR

Q.3(A) Write a detail note **De'Morgans Theorem.** **[06]**

(B) Simplify Boolean expression and draw circuit. **A.B+C.D+A.B+C.D** **[04]**

Q.4(A) Explain **8X3 Encoder.** **[06]**

(B) Define **K-Map.** Explain with proper example how it is used to reduced equation. **[04]**

OR

Q.4(A) Explain **3X8 Decoder.** **[06]**

(B) Define **K-Map.** Also explain **PAIR** in context of same. **[04]**

Q.5(A) Write a detail note on **4-bit Binary Adder** with Circuit Diagram. **[06]**

(B) Explain working of **Half Adder.** **[04]**

OR

Q.5(A) Write a detail note on **4-bit Binary Substractor** with Circuit Diagram. **[06]**

(B) Explain working of **Full Adder.** **[04]**

Q.6 Write a detail note on **Shift Registers.** Also discuss **Counters.** **[10]**

OR

Q.6 Define **FlipFlop.** Also discuss **D Latches (clocked and unclocked).** **[10]**

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